Yuanshuo Du

GitHub YuanshuoDu ▪ LinkedIn yuanshuodu ▪ Email duyuanshuo@gmail.com
Tel +353 89 454 6468 — Location Dublin, Ireland

Summary

Computer Vision and ML Engineering specialist with an MSc in Computer Science and deep learning coursework grounding. Hands-on expertise in CNN-based computer vision (ResNet50, R-CNN) and production ML system deployment. Daily practitioner of AI tools (Claude, RAG, n8n) for workflow automation, with a proven track record in Agile delivery.

Skills

ML / Deep Learning: CNN, ResNet50, R-CNN, Deep Learning, TensorFlow, PyTorch, scikit-learn, Python

Programming: Python, Go, Node.js, SQL, R

Data Engineering: PostgreSQL, MongoDB, Apache Spark, Data Visualization

AI Engineering: RAG, Claude API, FastAPI, Prompt Engineering, Docker, GitBook

Languages: Chinese (native), English (professional), Hungarian (conversational)

Technical Projects

Smart Parking Space Detection System — MSc CS Dissertation (Sep – Dec 2024)

CNN • ResNet50 • R-CNN • TensorFlow • Python • PostgreSQL

- Designed and trained CNN models (ResNet50, R-CNN) on satellite/aerial imagery to detect real-time parking space availability
- Achieved strong detection accuracy on test dataset using transfer learning from ResNet50 pretrained weights
- Co-authored and published research paper at *EduLearn25* Conference Proceedings
- Built backend services in Go and Node.js; designed PostgreSQL schema for real-time parking availability data

AI-Powered Q&A Platform — Mar 2024 – Sep 2024

RAG • Claude API • FastAPI • PostgreSQL • pgvector

- Built Retrieval-Augmented Generation (RAG) pipeline using Claude API and FastAPI to answer user queries over large document corpora
- Implemented document chunking, embedding storage in PostgreSQL (pgvector), and semantic search retrieval pipeline
- Authored engineering documentation in GitBook for cross-team collaboration and onboarding
- Deployed scalable backend services on cloud infrastructure with Docker containerization

GIS for Public Amenities in Urban Areas — 2024

GIS • Python • R • SQL • Data Visualization

- Developed spatial analysis pipeline for identifying and visualizing public amenity distribution in urban areas using GIS
- Processed and analyzed large urban datasets; produced choropleth maps and accessibility metrics using Python and R
- Published as peer-reviewed paper in *EduLearn25* Conference Proceedings

Professional Experience

Software Engineer (Intern) – AI/ML — Mar 2024 – Oct 2024

- Built and deployed Flask/FastAPI microservices for ML model inference endpoints on cloud infrastructure
- Instrumented model inference endpoints with Prometheus metrics; reduced monitoring setup time by 60% for new services
- Collaborated in Agile sprints using Jira; participated in code reviews and CI/CD pipeline maintenance on GitHub Actions
- Containerized AI/ML inference services with Docker; authored deployment runbooks for production environments

Education

MSc Computer Science — Technological University Dublin — Sep 2023 – Dec 2024

Deep Learning, Systems Architecture, Web App Architectures, Agile Methods

MSc Management (International Business) — University of Sheffield — Jun 2019 – Mar 2021

Agile Project Management, Accounting & Financial Management

Publications

Transfer Learning Approaches for Parking Space Detection from Satellite Imagery – EduLearn25 Conference Proceedings (Jul 2025)

The Development of a GIS for Public Amenities in Urban Areas – EduLearn25 Conference Proceedings (Jul 2025)